```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n = 5;
    double x[5] = {1, 2, 3, 4, 5};
    double y[5] = {1, 8, 27, 64, 125};
    double d[5][5] = {};

    for (int i = 0; i < n; i++)
        d[i][0] = y[i];

    for (int j = 1; j < n; j++) {
        for (int i = 0; i < n - j; i++) {
            d[i][j] = d[i + 1][j - 1] - d[i][j - 1];
        }
    }

    cout << "Difference Table\n";
    cout << fixed << setprecision(2);

    for (int i = 0; i < n; i++) {
        cout << x[i] << "\t";
        for (int j = 0; j < n - i; j++)
            cout << d[i][j] << "\t";
        cout << endl;
    }

    return 0;
}
```

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n = 5;
    double x[5] = {1, 2, 3, 4, 5};
    double y[5] = {1, 8, 27, 64, 125};
    double d[5][5] = {};

    for (int i = 0; i < n; i++)
        d[i][0] = y[i];

    for (int j = 1; j < n; j++)
        for (int i = 0; i < n - j; i++)
            d[i][j] = d[i + 1][j - 1] - d[i][j - 1];

    double xp = 1.7;
    double h = x[1] - x[0];
    double u = (xp - x[0]) / h;
    double result = d[0][0];
    double term = 1;

    for (int j = 1; j < n; j++) {
        term *= (u - (j - 1)) / j;
        result += term * d[0][j];
    }

    cout << fixed << setprecision(5);
    cout << "y(1.7) = " << result << endl;

    return 0;
}
```

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n = 5;
    double x[5] = {1, 2, 3, 4, 5};
    double y[5] = {1, 8, 27, 64, 125};
    double d[5][5] = {};

    for (int i = 0; i < n; i++)
        d[i][0] = y[i];

    for (int j = 1; j < n; j++)
        for (int i = 0; i < n - j; i++)
            d[i][j] = d[i + 1][j - 1] - d[i][j - 1];

    double xp = 4.7;
    double h = x[1] - x[0];
    double u = (xp - x[n - 1]) / h;
    double result = y[n - 1];
    double term = 1;

    for (int j = 1; j < n; j++) {
        term *= (u + j - 1) / j;
        result += term * d[n - j - 1][j];
    }

    cout << fixed << setprecision(5);
    cout << "y(4.7) = " << result << endl;

    return 0;
}
```

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n = 5;
    double y[5] = {1, 8, 27, 64, 125};
    double x[5] = {1, 2, 3, 4, 5};
    double yp = 85;
    double result = 0;

    for (int i = 0; i < n; i++) {
        double term = x[i];

        for (int j = 0; j < n; j++) {
            if (i != j)
                term *= (yp - y[j]) / (y[i] - y[j]);
        }

        result += term;
    }

    cout << fixed << setprecision(5);
    cout << "x when f(x) = 85: " << result << endl;

    return 0;
}
```

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n = 5;
    double x[5] = {1, 3, 4, 6, 10};
    double y[5] = {0, 18, 58, 190, 920};
    double d[5][5] = {};

    for (int i = 0; i < n; i++)
        d[i][0] = y[i];

    // Divided difference table
    for (int j = 1; j < n; j++) {
        for (int i = 0; i < n - j; i++) {
            d[i][j] =
                (d[i + 1][j - 1] - d[i][j - 1])
                / (x[i + j] - x[i]);
        }
    }

    cout << fixed << setprecision(4);
    cout << "Divided Difference Table:\n";

    for (int i = 0; i < n; i++) {
        cout << x[i] << "\t";
        for (int j = 0; j < n - i; j++)
            cout << d[i][j] << "\t";
        cout << endl;
    }

    double xp = 2.7;
    double result = d[0][0];
    double product = 1;

    for (int j = 1; j < n; j++) {
        product *= (xp - x[j - 1]);
        result += product * d[0][j];
    }

    cout << setprecision(5);
    cout << "f(2.7) = " << result << endl;

    return 0;
}
```